Journal of Environmental Psychology 52 (2017) 43–55

Contents lists available at ScienceDirect

Journal of Environmental Psychology

journal homepage: www.elsevier.com/locate/jep

Warm glow vs. altruistic values: How important is intrinsic emotional reward in proenvironmental behavior?

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ARTICLE INFO

Article history:
Received 3 May 2016
Received in revised form
13 April 2017
Accepted 31 May 2017
Available online 1 June 2017

Keywords:
Warm glow
Altruistic values
Proenvironmental behavior
Climate protection

ABSTRACT

This research addresses the role of warm glow as an antecedent of proenvironmental behavior in a comparative study of the behavioral effects of warm glow and altruistic personality traits and values. So far the influences of altruism and warm glow have not been analyzed simultaneously. Two online surveys with representative population samples show that warm glow has a stronger influence on proenvironmental intentions than altruism. However, the study also provides a process explanation for the decreased influence of altruism when warm glow is introduced into the model. Results show that warm glow mediates the effects of altruism, but that introducing warm glow together with altruism also explains additional variance of proenvironmental behavior, apart from the indirect effect of altruism. Further findings support a reinforcing mechanism by which warm glow strengthens the effect of prior proenvironmental behavior on future intention. Implications for the promotion of proenvironmental behavior are discussed.

1. Introduction

Do individuals behave prosocially because they are driven by selfless and altruistic values, or can prosocial behavior also be more instrumental and motivated by experiences of self-gratifying emotional reward? The current study tries to answer this question for the case of proenvironmental behavior.

Altruistic values have been conceptualized as part of a personal value structure or overall guiding principle that motivates individuals to contribute to the wellbeing of others or of society as a whole (Schwartz, 1972; Stern, Dietz, Kalof, & Guagnano, 1995). A stream of empirical evidence has supported altruistic values specifically as an antecedent of proenvironmental behavior (Bamberg, 2003; Berenguer, 2007; Stern & Dietz, 1994). Some researchers have however questioned the existence of 'true' altruism, that is, behavior motivated by authentically selfless motives, and proposed alternative accounts for prosocial motivation (Batson, 1987; Cialdini, Brown, Lewis, Luce, & Neuberg, 1997; Griskevicius, Tybur, & Van den Bergh, 2010). Andreoni (1989, 1990) showed that individuals may experience psychological benefits from prosocial behavior, which he termed "warm glow of giving". Warm glow benefits have also been shown to motivate proenvironmental behavior, since individuals may experience a feeling of moral satisfaction derived from their contribution to the common good environment (Kahneman & Knetsch, 1992; Nunes & Schokkaert, 2003; Ritov & Kahneman, 1997).

However, so far, the effects of altruism and warm glow have not been assessed in the same study and analyzed simultaneously. The present research addresses this gap in the literature, while also making theoretical propositions with regard to the behavioral antecedents and consequences of warm glow. The contribution of this research is threefold. First, this research compares the effects of altruistic values and warm glow as drivers of environmental behavior. Second, a process explanation for altruism effects is proposed and tested, with warm glow mediating behavioral

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http://dx.doi.org/10.1016/j.jenvp.2017.05.006
0272-4944/

influences of altruism. Third, this research studies whether warm glow can both drive and reinforce proenvironmental behavior.

These contributions are based on robust findings across two field studies. The first study assessed the influence of altruism and warm glow on two environmental behaviors with 600 individuals drawn form a representative panel of the Australian population. The second study, in addition to reassessing the influences of altruism and warm glow, further addressed the reinforcing influence of warm glow by comparing 300 Australian residential customers of green electricity with a representative sample of 300 Australians with standard home electricity.

2. The role of altruism in proenvironmental behavior

Altruism involves acting to increase the welfare of others incurring personal costs but lacking personal gains (Batson, 1994). As with most prosocial behavior, proenvironmental behavior has inherent characteristics of altruism and can be construed as such (Griskevicius et al., 2010). Since the impact of an individual's proenvironmental behavior on its own marginal welfare is rather negligible, the individual acts for the benefit of the common good or future generations, incurring actual costs but lacking an increase in personal welfare.

Rushton, Chrisjohn, and Fekken (1981) suggested a broad base for altruism derived from an "altruistic personality" and proposes the existence of altruistic personality traits. With the term "altruistic traits" these authors referred to enduring personality characteristics which are revealed in particular behavior patterns.

Altruism has also been addressed from a human values perspective. Personal values are conceptualized as desirable, *trans*-situational goals that serve as guiding principles in people's lives (Schwartz, 1992). Behavioral research stretching over several decades has confirmed human values as important drivers of proenvironmental behaviors. Linking proenvironmental behavior to particular values, research has drawn on the value measures developed in cross-national research by Schwartz (1992), based in turn on Schwartz's (1972) norm-activation theory of altruism. Several studies have found that self-transcendence or altruistic values (i.e., universalism and benevolence in Schwartz's value classification scheme) are stronger among people who engage in proenvironmental activities (e.g, Schwartz, 1992; Stern et al., 1995). Proenvironmental attitudes and behaviors have been shown to be positively related to altruistic or self-transcendence values, while negatively related to egoistic or self-enhancement values (Gärling, Fujii, Gärling, & Jakobsson, 2003; Hansla, Gamble, Juliusson, & Gärling, 2008a; Kaiser, Hübner, & Bogner, 2005; Nordlund & Garvill, 2002; Stern & Dietz, 1994). Also the influences of environmental-altruistic values have been confirmed in several studies (Stern, Dietz, & Kalof, 1993; Stern et al., 1995). Further research centered on climate protection and green energy consumption found that altruistic values contributed to explaining participation in a green energy program (Clark, Kotchen, & Moore, 2003) and that willingness to pay for green electricity related to self-transcendence value types (Hansla, Gamble, Juliusson, & Gärling, 2008b).

3. Is behavior ever truly altruistic?

Notwithstanding findings on the effects of altruistic values, there is still an ongoing scientific debate over whether individuals are ever truly altruistic (Batson, 1987). It has been suggested that prosocial motives are egoistic when the ultimate goal is to increase one's own welfare, and altruistic when it is to increase others' welfare (Batson, Fultz, & Schoenrade, 1987). A number of researchers, such as Cialdini et al. (1997), propose non-altruistic accounts of prosocial motivation, questioning pure altruism. A common view is that human's behavioral motivations are essentially selfish. Behavior that appears to be altruistic can actually have egoistic motives such as reputation seeking or reduction of distress and aversive arousal, evoked as a consequence of being aware of other individuals' needs or suffering (Batson, 1987; Cialdini et al., 1987).

4. Warm glow and proenvironmental behavior

4.1. Operationalizing warm glow

Pleasure seeking and pursuing happiness may constitute important motives for prosocial and proenvironmental behavior. Isen (1970) found that the potential helper's positive affective state determines helpfulness ("people who feel good themselves are more likely to help others") and introduced the term "warm glow" to refer to the emotional experience involved. Warm glow may increase the probability of engaging in prosocial behavior. This "warm glow effect" has received broad empirical support. Research has shown that experiencing feelings of pleasure can significantly increase an individual's likelihood of engaging in charitable behavior (e.g., Isen & Levin, 1972; Levin & Isen, 1975) and can induce helping behavior (Cunningham, Steinberg, & Grev, 1980). Analyzing the case of blood donations, Allen, Machleit, and Schultz Kleine (1992) showed that emotional gratifications such as the feeling of joy accompanied donation behavior. Duclos and Barasch (2014) found that the belief that helping others contributed to personal happiness had a significant influence on prosocial behaviors.

A further development of warm glow theory has originated from an economic perspective of behavior. To account for the observed incongruence of donation behavior with classical economic theory, Andreoni (1989, 1990) suggested that when people make donations to public goods, instead of being motivated by the increase in value of the common good, they rather experience a direct, personal benefit arising from the contribution itself. This personal benefit, which Andreoni called the "warm glow of giving", is of purely psychological character and arises from the moral satisfaction that an individual derives from the act of giving.

Most previous empirical research has addressed warm glow indirectly as an additional component of an individual's utility function, without measuring it explicitly. Kahneman and Knetsch (1992) developed a first empirical approach to the direct measurement of warm glow, focusing, however, exclusively on the degree of satisfaction an individual would receive from contributing to environmental causes. In their experiment, the individual willingness to contribute was predicted by ratings of the moral satisfaction associated with the contribution. In fact, the measurement item used by Kahnemann and Knetsch was nearly identical to post-consumption satisfaction measures commonly used in the customer satisfaction literature (e.g., Mano & Oliver, 1993; Westbrook & Oliver, 1991). Table 1 provides an overview over assessments and measures of warm glow in prior studies and their problems.

The warm glow operationalization proposed in the current study is in part based on Isen's (1970) and Isen and Levin's (1972) concept of warm glow as an affective-emotional state, and captures the feeling of wellbeing related to the contribution to a good cause. Following the cognitive appraisal perspective (Bagozzi, Gopinath, & Nyer, 1999; Roseman, 1991), warm glow feelings may result from the cognitive appraisal of the contribution of prosocial and proenvironmental behavior to the common good. Based on this argumentation, and adding to the perspective of Kahneman and Knetsch (1992), we propose assessing warm glow as a positive affective response, in terms of Roseman's (1991) definition of an "appetitive mental state" or "presence of reward", such as feeling happy, pleased, satisfied or content. Compared to existing measures of affect such as Watson and Tellegen's (1985) Positive and Negative Affect Schedule (PANAS), developed to address unspecific situational emotion or mood states, a warm glow measure ought to specifically assess positive affective reward derived from the cognitive appraisal of the contribution of the individual's prosocial and proenvironmental behavior to the common good.

In the present research, warm glow is therefore operationalized as *an emotional construct of feelings of pleasure and satisfaction derived from the cognitive appraisal of contributing to the wellbeing of society, and, specifically, to environmental protection*. The proposed operationalization of warm glow improves on prior approaches insofar as it goes beyond Kahneman and Knetsch's (1992) satisfaction experience, resolves limitations of existing alternative measures, while capturing positive affect as the core dimension of intrinsic emotional reward.

4.2. Warm glow as driver of proenvironmental behavior

Inspired by Andreoni's (1989, 1990) research, Kahneman and Knetsch (1992) applied warm glow theory to explaining willingness to contribute to environmental quality. Their findings showed that the stated willingness to contribute with monetary donations reflects the willingness to pay for the moral satisfaction derived from contributing to the public good environment, rather than the economic value of improvement in environmental quality. Individuals seem to be willing to pay a certain amount to acquire warm glow from contributing to environmental protection. Subsequent research provided supportive evidence for the influence of warm glow on proenvironmental behavior (e.g., Irwin & Spira, 1997; Kahneman & Ritov, 1994; Nunes & Schokkaert, 2003; Ritov & Kahneman, 1997).

Warm glow theory has also been applied to particular environmental behaviors related to climate protection and reducing greenhouse-gas emissions such as energy saving, green electricity adoption and purchasing products with a reduced carbon footprint. These behaviors constitute a contribution to a public good, for an improvement in climate quality is consumed jointly by all individuals and it is not possible to exclude certain individuals from consuming it. Benefits of climate protection are consumed by the society as a whole, while additional costs such as premium prices paid are absorbed by the individual. Only few studies have addressed the relationship between warm glow and willingness to behave in favor of climate protection empirically. Addressing the purchase of products with reduced carbon footprints, Irwin and Spira (1997) showed that the emotional and moral benefit derived from environmental product attributes explained the valuation of such products. Roe et al.'s (2002) stated preferences experiment confirmed warm glow as part of the perceived utility of contributing to a green electricity program. Further evidence for the warm glow effect in green electricity adoption has been provide by Ek and Söderholm (2008) and Hartmann and Apaolaza-Ibáñez (2012). While the warm glow effect in proenvironmental behavior has been supported in the literature, the measure of this construct differed from the present operationalization. Thus we propose that warm glow experiences, operationalized in the proposed manner as a directly measured emotional reward, should motivate the intention to engage in proenvironmental behavior such as, for instance, switching to green electricity for the sake of climate protection:

H1. Warm glow, operationalized as an anticipated positive affective experience derived from proenvironmental behavior such as contributing to climate protection, has a positive influence on proenvironmental intentions.

5. Warm glow versus altruism: what drives proenvironmental behavior?

Most prior literature addresses altruism and warm glow as

Table 1
Assessment and measures of warm glow in prior research.

Study	Operationalization	Problem
Andreoni (1989, 1990), Kotchen (2005), Kotchen and Moore (2007), Nunes and Schokkaert (2003)	Implicit variable in utility function.	Not measured, but inferred from behavioral reactions.
Clark et al. (2003)	"I take satisfaction in participating in this program, regardless of its environmental effects"	Cognitive prompting participants with the idea that they engaged in environmental behavior regardless its effects confounds the warm glow effect. This may explain why participants ranked warm glow least in importance among proenvironmental motives.
Duclos and Barasch (2014)	Extent to which participants believed that donating money can promote their own personal happiness.	Measures happiness, i.e., only one aspect of the emotional experience of warm glow.
Dunn, Aknin, and Norton (2008)	Increase in general happiness after spending on others (gifts or donations).	Measures happiness, only one aspect of the emotional experience of warm glow.
Hartmann and Apaolaza-Ibáñez (2012)	"Clients of Brand X can feel good because they help to protect the environment.", "With Brand X, I have the feeling of contributing to the well-being of humanity and nature.", "Clients of Brand X can feel better because they don't harm the environment."	Ad-hoc measure, no scale development procedure. Narrow application to brand benefits.
Isen (1970); Isen and Levin (1972); Levin and Isen (1975)	Induced with experimental manipulation (giving a monetary incentive to participants), based on the assumption that if something good happens to individuals they are more inclined to help others.	Not measured, but inferred from experimental manipulation.
Kahneman and Knetsch (1992)	"Indicate the degree of satisfaction you would receive from contributing to each cause on a scale from 0 to 10 with 0 indicating no satisfaction at all and 10 indicating a great deal of personal satisfaction."	Measures satisfaction, i.e., only one aspect of the emotional experience of warm glow.
Strahilevitz and Myers (1998)	Intrinsic, interpreted as a feeling of pleasure experienced as a consequence of purchasing a consumer product bundled with charity incentives.	Not measured, but inferred from behavioral reactions.

alternative explanations of prosocial behavior. As discussed previously, some authors question "pure altruism" (Batson, 1987; Cialdini et al., 1997; Griskevicius et al., 2010) and propose warm glow as an alternative, egoistic motive for seemingly "selfless" prosocial and proenvironmental behavior (e.g., Andreoni, 1989, 1990; Kahneman & Knetsch, 1992). However, so far the behavioral effects of warm glow and altruism have not been jointly analyzed and compared by means of explicit measures. It remains yet unknown whether altruism or warm glow has a stronger influence on proenvironmental behavior. Introducing warm glow into a model explaining environmental behavior based on altruistic values may explain additional variance beyond the influences of altruism and/or displace its effects.

RQ1: Do altruistic values and personality traits or warm glow emotional reward have a stronger influence on proenvironmental intentions?

In a parallel effect model, if warm glow would displace a significant part of the behavioral effect of altruism or even altogether substitute the effect of altruism, this may imply that indeed warm glow was a more important antecedent of proenvironmental behavior. On the other hand, warm glow effects may instead provide a complementary explanation to altruism. Indeed, a mediation model may provide a process explanation of behavioral effects of altruism. Altruistic behavior has been shown to induce warm glow (e.g., Andreoni, 1990; Kahneman & Knetsch, 1992; Kahneman & Ritov, 1994). Altruistic values may therefore enhance the anticipated and/or post-hoc experience of warm glow. Warm glow, in turn, may mediate altruism effects on proenvironmental behavior.

From a theoretical perspective, the direction of the potential mediation is clearly defined since the opposite effect is unreasonable: expected or experienced warm glow will not change any individual's value structure. Experiencing warm glow may therefore positively relate to altruistic values and personality traits. Furthermore, it seems probable that holding altruistic values would enhance the experience of warm glow and that the anticipated emotional reward from this experience constituted the ultimate motive for engaging in prosocial and proenvironmental behavior. The effect of altruism would consequently be at least partially indirect, mediated by anticipated warm glow experiences.

H2. Altruistic values and personality traits have an indirect positive influence on proenvironmental intentions, mediated by warm glow experiences.

Fig. 1 illustrates the two alternative models describing the influences of altruism and warm glow on proenvironmental behavior.

6. Warm glow as a reinforcing process in proenvironmental behavior

Experiencing warm glow as a consequence of proenvironmental action may also lead to learning effects affecting future behavior. As suggested in H1, anticipation of warm glow as affective reward may constitute an important motive inducing proenvironmental behavior. In addition, warm glow should be experienced once the individual has actually engaged in proenvironmental behavior. Several studies have found warm glow as a consequence of prosocial and proenvironmental behavior (e.g., Dunn et al., 2008; Menges, Schroeder, & Traub, 2005; Strahilevitz & Myers, 1998). Hence, anticipated warm glow reward can be considered an antecedent and experienced warm glow an outcome of proenvironmental behavior. Thus, if warm glow shows effects as driver and as outcome of behavior, it describes a reinforcing mechanism. Experiencing warm glow derived from actual proenvironmental behavior should drive future proenvironmental behavior.

H3. Warm glow, operationalized as an anticipated positive affective experience derived from proenvironmental behavior such as contributing to climate protection, mediates the relationship between prior proenvironmental behavior and future proenvironmental intentions.

Fig. 2 depicts the proposed reinforcing mechanism.

7. Study 1: effects of altruism and warm glow on proenvironmental intentions

7.1. Participants and procedure

Study one was conducted to investigate the simultaneous effect of altruism and warm glow on proenvironmental intentions, verifying the relationships proposed in hypotheses 1 and 2, and addressing the research question. A sample of 600 individuals was drawn from a representative online-panel of the Australian population recruited by commercial online-panel provider Pureprofile. Subjects were selected by a quota based random criterion to match the population in terms of age, sex, income, education and geographic distribution, and received a monetary compensation for participating in the study (see Appendix 1 for sample characteristics).

Participants were presented with an online questionnaire on environmental issues without previous exposure to any information or stimuli. The study addressed two exemplary proenvironmental behaviors related to climate protection: intention to sign up for a residential green electricity contract and support for a tax on carbon dioxide emissions. Both studies were based in Australia because a comprehensive governmental green electricity certification program was implemented and, at the same time, there was an ongoing political debate regarding a tax on carbon emissions. Participants could therefore be expected to be familiar with the topics green electricity and taxing carbon emissions.

7.2. Measurement

The questionnaire comprised measures of intention to engage in proenvironmental behaviors related to climate protection, a measure of warm glow experiences derived from contributing to environmental protection, and items measuring altruistic personality traits and values.

Proenvironmental intentions were addressed through two behaviors related to climate protection: *intention to sign up for a residential green electricity contract* and *support for a tax on carbon dioxide emissions*. To assess *purchase intention*, participants were presented with information on two alternative electricity contracts: i) standard electricity plan, generated from non-renewable energy sources, at an average monthly cost of AUD$102; ii) greenpower plan, providing certified green electricity from renewable energy sources like wind, solar and hydro-energy at an average monthly cost of AUD$142. Price information was derived by averaging actual online offers of different Australian electricity providers for an average level of electricity consumption. Subjects were asked to rate on a five point scale the likelihood of choosing the certified Greenpower plan, if they had to decide about a new electricity contract for their home: *definitely will not* = 1, *probably will not* = 2, *might/might not* = 3, *probably will* = 4, *definitely will* = 5 ($M = 2.77$, $SD = 1.14$). Subsequently, *support for a tax on carbon*

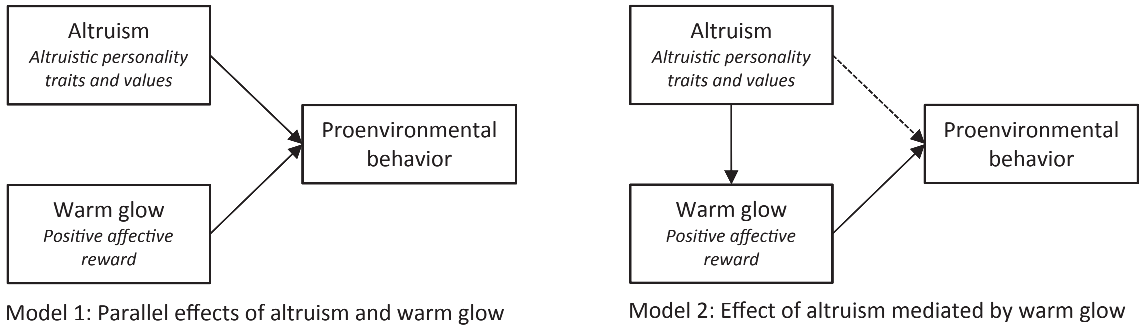

Fig. 1. Alternative models of parallel and mediated effects of altruism and warm glow on proenvironmental behavior.

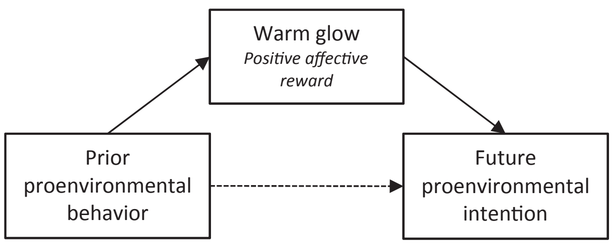

Fig. 2. Reinforcing influence of warm glow: indirect effect of prior proenvironmental behavior on future proenvironmental intentions mediated by warm glow.

dioxide emissions was assessed by asking participants how likely they would support such a tax in a hypothetical referendum (popular vote) about this issue in Australia. Likelihood of tax support was rated on a five-point scale with identical scaling labels as the green power purchase intention variable (M = 2.29, SD = 1.47).

Warm glow was measured in line with the proposed operationalization as an emotional construct construed of feelings of pleasure and satisfaction derived from the appraisal of contributing to the wellbeing of society, and, in the specific context of the empirical study, to climate protection. Consistent with accepted typologies of emotional items related to positive affective reward (e.g., Bagozzi et al., 1999; Edell & Burke, 1987; Watson & Tellegen, 1985), warm glow might be addressed as feeling, happy, pleased satisfied and content.

However, since the warm glow measure assesses positive affective reward specifically derived from the individual's cognitive appraisal of the contribution of his or her prosocial and proenvironmental behavior to the common good, the measurement scale needs to be more specific than existing affect measures (e.g., Positive and Negative Affect Schedule (PANAS); Watson & Tellegen, 1985) which measure unspecific situational emotion or mood states. To correctly assess and differentiate warm glow experiences from other emotional responses, the measurement items composing the scale should explicitly relate to the specific proenvironmental behavior potentially evoking warm glow. That is, the measure ought to establish a connection between the emotions elicited and the particular proenvironmental behavior in question.

A series of individual and group interviews were carried out to explore affective warm glow experiences, among other environmental motives and attitudes. Besides several group interviews with undergraduate and graduate students, six in-depth interviews were conducted with individuals who had signed up for green electricity contracts at their homes, paying a premium price for electricity from renewable sources. These subjects were traced by a professional market research institute, received a monetary compensation for participating in the interviews, and were selected from six different geographic zones of the city of Melbourne, Australia. Interviewees were prompted to elaborate on their feelings that were elicited by the fact that they had signed up for green electricity contracts in their homes. All interviewees revealed positive affective experiences derived from their contribution with statements such as "*I feel good (…), it's my little contribution*", "*I'm really very happy that I can do something*", "*I feel good that I am doing what I can*", expressing "*feeling happy and content making a bit of a difference*" and experiencing "*a feeling of pride and satisfaction in doing something that helps*" (see Appendix 2 for excerpts of the in-depth interviews). Based both on the proposed operationalization of warm glow and the exploratory study, four measurement items were developed (Table 2). Subjects rated their agreement or disagreement with these statements on five-point Likert-type agreement scales (*strongly disagree* = 1 to *strongly agree* = 5). The introducing question asked participants *how contributing to reducing global warming and climate change would make them feel (by switching to green electricity, saving energy, supporting climate policies, etc.)*.

The validation of the scale provided overall satisfactory results. The Kaiser-Meyer-Olkin measure of sampling adequacy was high with 0.87 and Bartlett's test of sphericity was highly significant ($p < 0.001$). The dimensionality of the scale was analyzed with principal component analysis. One single factor was extracted with eigenvalue 3.68, 92% of explained variance, and factor loadings ranging from 0.95 to 0.98. Cronbach's alpha of the scale was 0.97, indicating high scale reliability. The *warm glow* measure was computed by averaging the ratings of the four items (M = 3.36, SD = 1.13).

Altruistic values were assessed with a measurement instrument developed by Stern et al. (1993), based on Schwartz's (1972, 1992) value measures, and validated in previous research (e.g. Steg, Dreijerink, & Abrahamse, 2005; Stern & Dietz, 1994; Stern et al., 1995). The scale was composed of eight value items, of which four represented self-transcendence altruistic values (*equality, a world at peace, social justice,* and *helpful*; M = 5.14, SD = 1.69, Cronbach's alpha = 0.89) and four environmental-altruistic values (*preventing pollution, respecting the earth, unity with nature,* and *protecting the environment*; M = 4.79, SD = 1.86, Cronbach's alpha = 0.96). Using Schwartz's (1992) rating scale, subjects were asked to rate the importance of these values as guiding principles in their lives on nine-point scales labeled *opposed to my values = -1, not important = 0, 1 (unlabeled), 2 (unlabeled), important = 3, 4 (unlabeled), 5 (unlabeled), very important = 6, extremely important = 7.*

Altruistic personality traits were assessed with Rushton, Chrisjohn, and Fekken's (1981) "*altruistic personality and self-*

Table 2
Confirmatory factor analyses of measurement constructs (Study 1 and 2).

Measurement constructs and indicators	Study 1					Study 2 – standard elec.					Study 2 – green elec.				
	B	SE	β	AVE	CR	B	SE	β	AVE	CR	B	SE	β	AVE	CR
Warm glow				0.91	0.98				0.90	0.98				0.84	0.96
Doing something about climate change gives me a pleasant feeling of personal satisfaction.	1.00	0.00	0.92			1.00	0.00	0.92			1.00	0.00	0.84		
Reducing my personal carbon emissions, I feel happy contributing to human wellbeing and the quality of the natural environment.	0.99	0.03	0.92			1.01	0.04	0.91			1.04	0.05	0.91		
Helping to prevent climate change, I feel pleased to be doing something good for our planet.	1.04	0.02	0.98			1.05	0.03	0.96			1.00	0.05	0.92		
Participating in climate protection makes me feel satisfied, giving something back to society and the environment.	1.05	0.02	0.96			1.05	0.03	0.97			1.04	0.05	0.92		
Altruistic personality traits				0.45	0.85				0.44	0.83				0.43	0.86
I have given money to a charity or environmental organization.	1.00	0.00	0.52			1.00	0.00	0.47			1.00	0.00	0.53		
I have given money to a stranger who needed it (or asked me for it).	0.92	0.11	0.50			0.89	0.17	0.42			1.54	0.20	0.70		
I have done volunteer work for a charity or environmental organization.	1.24	0.13	0.56			1.10	0.21	0.44			1.35	0.20	0.54		
I have allowed someone to go ahead of me in a queue (in the supermarket, etc.).	1.05	0.10	0.72			1.08	0.16	0.66			1.10	0.14	0.71		
I have offered my seat on a bus, tram or train to a stranger who was standing.	1.17	0.11	0.72			1.54	0.22	0.84			1.18	0.15	0.71		
Self-transcendence altruistic values				0.76	0.94				0.67	0.91				0.79	0.95
Equality: equal opportunity for all	1.00	0.00	0.70			1.00	0.00	0.70			1.00	0.00	0.75		
A world at peace: free of war and conflict	1.07	0.06	0.78			0.96	0.08	0.72			0.94	0.06	0.85		
Social justice: care for the weak	1.34	0.07	0.93			1.23	0.09	0.90			1.23	0.07	0.95		
Helpful: working for the welfare of others	1.33	0.07	0.89			1.17	0.09	0.82			1.16	0.08	0.86		
Environmental-altruistic values				0.90	0.98				0.87	0.97				0.87	0.97
Preventing pollution: protecting natural resources	1.00	0.00	0.89			1.00	0.00	0.88			1.00	0.00	0.88		
Respecting the earth: harmony with other species	1.06	0.03	0.96			1.05	0.04	0.95			1.07	0.04	0.96		
Unity with nature: fitting into nature	1.08	0.03	0.94			1.04	0.04	0.92			1.12	0.05	0.90		
Protecting the environment: preserving nature	1.04	0.03	0.94			1.00	0.04	0.92			1.07	0.04	0.95		
Kaiser-Meyer-Olkin*	683.43–3287.27					267.90–1626.19					384.75–1326.00				
Bartlett's Test of Sphericity*	(p < 0.001)					(p < 0.001)					(p < 0.001)				
	0.72–0.87					0.68–0.86					0.77–0.86				
Model fit	CFI = 0.96, NFI = 0.95, RMSEA = 0.07, CMIN/DF = 4.53					CFI = 0.96, NFI = 0.93, RMSEA = 0.05, CMIN/DF = 2.57									

Note. * Value range; NFI = normed fit index, CFI = comparative fit index, RMSEA = root mean square error of approximation.

report altruism scale" based on the perspective of altruism as a personal trait or "altruistic personality". Participants had to indicate how often they had carried out five prosocial behaviors/activities in the past (incurring actual costs but without an increase in personal welfare, such as donations to charity or persons in need, volunteering, offering strangers to go ahead of oneself in a queue or to take one's seat in a crowded bus) on five-point scales labeled *never* = 1 to *very often* = 5 (Table 2). Factor loadings were acceptable ranging from 0.65 to 0.75, and 49% of explained variance. Cronbach's alpha of the scale was 0.74, supporting adequate reliability ($M = 3.19$, $SD = 0.83$). Table 2 shows measurement constructs and indicators.

Confirmatory factor analysis with AMOS 20's maximum likelihood estimation confirmed an acceptable fit of the measurement model (Table 2) with normed fit index (NFI) = 0.95, comparative fit index (CFI) = 0.96 (Bentler, 1990), and root mean square error of approximation (RMSEA) = 0.07 (McDonald & Ho, 2002; Steiger & Lind, 1980; Steiger, 1990). Convergent and discriminant validity of the measurement scales was assessed according to Fornell and Larcker (1981). The test proved appropriate scale validity for nearly all constructs with variance extracted and composite reliability exceeding their respective recommended minimum thresholds of 0.50 and 0.70 (Hair, Anderson, Tatham, & Black, 1998). The only exception was the *altruistic personality traits* variable where variance extracted (0.45) was somewhat lower than the minimum recommended value; but as all other indicators were satisfactory and the scale has been frequently employed in the literature, the measure was kept. Furthermore, the discriminant validity of all constructs of the measurement model was confirmed since the square of construct correlations did not exceed variance explained by the corresponding correlated constructs in any case (Anderson & Gerbing, 1988; Fornell & Larcker, 1981). The correlations between all latent constructs of the measurement model ranged 0.24 to 0.76.

7.3. Results

Correlation analysis (Table 3) revealed a significant positive correlation of both environmental behaviors *intention to purchase green electricity* (PI) and *tax support* (TS) with the variables assessing altruism and warm glow. Stepwise multiple regression analyses with the enter method were conducted to address the hypothesized relationships and the research question (Table 4).

Regarding assumptions testing, the normal probability (P-P) plot of standardized regression residuals was nearly overlapping the diagonal, confirming approximately normal distributed errors. Also, with detection-tolerance values ranging 0.38 to 0.80 and variance inflation factors (VIF) ranging 1.3 to 2.6, indicators for multicollinearity were respectively well above and below recommended thresholds for tolerance > 0.10 to 0.20 and VIF < 5 to 10. Missing values ranged 18 (3.0%) to 35 (5.8%) and were excluded listwise ($N = 600$). Little's MCAR test confirmed randomness of missing values (chi-square = 85.17, $p = 0.27$).

When warm glow was introduced into the regression model together with the variables assessing altruistic behavioral tendencies and altruistic values (PI8, TS8), the influence of altruism turned non-significant with exception of the more specific environmentally oriented environmental values construct, while the warm glow effect is significant ($\beta_{PI} = 0.57$ and $\beta_{TS} = 0.50$, $p < 0.001$). The comparison between the full models ($R^2_{PI8} = 0.42$ and $R^2_{TS8} = 0.32$) with the respective models excluding warm glow ($R^2_{PI7} = 0.18$ and $R^2_{TS7} = 0.13$) showed that introducing warm glow increased explained variance significantly ($\Delta F_{PI} = 213.28$ and $\Delta F_{TS} = 149.78$, $p < 0.001$), highlighting the important explanatory

Table 3
Variable correlations (Study 1).

	PI	TS	AT	AV	EV
Green purchase intention (PI)	1.00				
Carbon tax support (TS)	0.52	1.00			
Altruistic personality traits (AT)	0.25	0.19	1.00		
Self-transcendence altruistic values (AV)	0.23	0.25	0.41	1.00	
Environmental-altruistic values (EV)	0.38	0.36	0.37	0.75	1.00
Warm glow (WG)	0.63	0.55	0.27	0.30	0.45

Note. All correlations $p < 0.001$.

Table 4
Effects of altruism and warm glow on environmental behaviors green electricity purchase intention (PI-models) and carbon tax support (TS-models): stepwise multiple regression model comparison (Study 1).

DV	Model	IV				R^2
		AT	AV	EV	WG	
PI	PI1	0.25^{***}				0.06
	PI2	0.07^{*}			0.61^{***}	0.40
	PI3		0.23^{***}			0.05
	PI4		0.05		0.61^{***}	0.40
	PI5			0.38^{***}		0.14
	PI6			0.13^{***}	0.56^{***}	0.41
	PI7	0.14^{**}	0.17^{**}	0.46^{***}		0.18
	PI8	0.05	0.08	0.17^{***}	0.57^{***}	0.42
TS	TS1	0.19^{***}				0.04
	TS2	0.05			0.55^{***}	0.31
	TS3		0.25^{***}			0.06
	TS4		0.10^{**}		0.52^{***}	0.31
	TS5			0.36^{***}		0.13
	TS6			0.14^{**}	0.49^{***}	0.32
	TS7	0.08^{a}	−0.06	0.36^{***}		0.13
	TS8	0.01	0.01	0.12^{*}	0.50^{***}	0.32

Note. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$, $^{a}p < 0.10$; standardized regression coefficients beta; multiple R-square; N = 600; missing values excluded listwise; DV = dependent variable; IV = independent variable; AT = altruistic personality traits; AV = self-transcendence altruistic values; EV = environmental-altruistic values; WG = warm glow.

power of warm glow as an antecedent of environmental behavior, as compared to altruism. The predictive utility of either altruism or warm glow was compared using Steiger's (1980) z test for non-independent correlations. For this test, the variance explained by a model introducing only warm glow was compared with the variance explained by a model based on the altruism variables (AT, AV and EV) excluding warm glow. The warm glow model explained significantly more variance in both dependent variables than did the alternative model ($Z_{PI} = 6.23$, $p < 0.001$ and $Z_{TS} = 5.04$, $p < 0.001$). Hence, the warm glow effect was significantly stronger than the influence of altruism and, in particular, warm glow had a stronger influence than environmental-altruistic values, the only significant altruism effect remaining in the full model (RQ1).

Results also supported the proposed mediation model with warm glow mediating the effects of altruism on proenvironmental behavior (H2). Table 4 shows the stepwise regression analyses of the effects of each of the altruism variables alone and introducing mediator warm glow (regression models PI1 to PI6 and TS1 to TS6). Introducing the mediator, the effect of altruism is significantly reduced (effects of altruistic personality traits (AT) and environmental-altruistic values (EV) on PI, and self-transcendence altruistic values (AV) and EV on TS) or even removed (effects of AV on PI and AT on TS).

Mediation analyses of indirect effects by both Sobel tests (Baron & Kenny, 1986; Sobel, 1982) and bias corrected bootstrap confidence intervals with 10,000 bootstrap samples (Hayes, 2013) confirmed indirect behavioral effects of altruism mediated by warm glow (Table 5). Consistent with the hypothesized pattern of relationships (H2), a significant indirect effect via warm glow on intention to switch to green electricity can be observed for altruistic personality traits ($b_{ind} = 0.24$, 95% Boot CI [0.17, 0.32], $Z = 6.46$, $p < 0.001$), self-transcendence altruistic values ($b_{ind} = 0.13$, 95% Boot CI [0.09; 0.17], $Z = 6.92$, $p < 0.001$) and environmental-altruistic values ($b_{ind} = 0.17$, 95% Boot CI [0.14; 0.20], $Z = 9.61$, $p < 0.001$). Results for influences on carbon tax support are nearly identical. The observed pattern of indirect effects contributes to explaining the decreased influence of altruism, once warm glow is introduced into the regression model. Altruistic values and personality traits may induce an individual to experience warm glow when engaging in environmental behavior and the anticipated affective reward may then constitute the final behavioral motive, rather than altruistic values. It is worth noting, however, that the role of warm glow is not limited to mediating altruism effects, since the warm glow variable on its own explains additional variance of proenvironmental behavior, apart from the indirect effect of altruism.

8. Study 2: altruism and warm glow of green vs. non-green individuals

Study 2 was designed to replicated the findings of study 1 and, in addition, to investigate the role of warm glow as a reinforcing mechanism for proenvironmental behavior as suggested in hypothesis 2 by comparing individuals who were signed into green electricity contracts with standard electricity clients. For this purpose, prior proenvironmental behavior was studied as an antecedent of warm glow that, in turn, influences future behavior. The two dependent variables were willingness to pay a price premium for green electricity (WTP) and, as in study 1, support for a carbon tax (TS).

8.1. Method

Three-hundred subjects signed into green electricity contracts were selected through two filter questions out of a representative random sample drawn from the Pureprofile panel of the Australian population. The filter questions assessed first, whether the individual was involved in the choice of the electricity provider at his or her home and, second, if his or her home was signed into a certified green electricity contract. Subjects were also filtered following a 50% quota on gender. Of the contacted individuals 8% had a green electricity contract. In addition, a representative sample of 300 individuals was drawn from the panel following the criteria of study 1 but discarding green electricity clients. Both samples are distinguished by the variable "green choice" (GC; standard electricity = 0; green electricity = 1).

All variables related to warm glow and altruistic personality traits and values were measured with identical measurement scales as in study 1. Data collected in the second study again supported scale validity of the *warm glow* measure. The Kaiser-Meyer-Olkin measure was 0.87 as in study 1 and Bartlett's test of sphericity was also significant ($p < 0.001$). Principal component analysis extracted one single factor with factor loadings ranging from 0.94 to 0.97 and explaining 92% of variance. Cronbach's alpha of the scale was 0.97, indicating high scale reliability. Multi-group confirmatory factor analysis with AMOS 20 indicated an acceptable fit of the measurement model (Table 2) in both the green and non-green group of participants. Convergent validity of the measurement scales was also supported in both groups with variance extracted and composite reliability exceeding their respective recommended minimum thresholds. Latent construct correlations ranged 0.17 to 0.68 in the green group and 0.34 to 0.75 in the non-

Table 5
Indirect effects of altruism on intention to purchase green electricity and tax support mediated by warm glow (Study 1).

DV	IV	Indirect Effect (B)	Boot SE	Boot LLCI	Boot ULCI	Z
PI	Altruistic personality traits (AT)	0.24	0.04	0.17	0.32	6.46***
	Self-transcendence altruistic values (AV)	0.13	0.02	0.09	0.17	6.92***
	Environmental-altruistic values (EV)	0.17	0.02	0.14	0.20	9.61***
TS	Altruistic personality traits (AT)	0.26	0.04	0.18	0.35	5.98***
	Self-transcendence altruistic values (AV)	0.13	0.02	0.09	0.18	6.36***
	Environmental-altruistic values (EV)	0.18	0.02	0.14	0.22	8.50***

Note. ***$p < 0.001$; 10,000 bootstrap samples for bias corrected bootstrap confidence intervals; DV = dependent variable; IV = independent variable; Boot SE = bootstrap standard error; Boot LLCI = bootstrap lower limit confidence interval; Boot ULCI = bootstrap upper limit confidence interval; PI = intention to purchase green electricity; TS = tax support.

green group. In none of the groups the square of construct correlations exceeded variance explained by the corresponding correlated constructs, confirming discriminant validity of all measures.

The proenvironmental intention *support for a carbon tax* was assessed as in study 1. Because the sample of study 2 included subjects already signed into green electricity contracts, the purchase intention variable of study 1 could not be used. Instead, *willingness to pay a premium price for green electricity* (WTP) was measured. Participants were asked the following question: "*If you had to decide about a new electricity contract for your home, how much more (in %) would you be willing to pay for green electricity instead of electricity from non-renewable sources.*" Variable mean values in the green and standard electricity sub-samples are presented in Table 6. Since assumption testing for the subsequent regression analyses suggested a logarithmic rather than linear function of WTP (see next section), we used ln(WTP) in the subsequent analyses.

8.2. Results

Consistent with results of study 1 and supporting H1, both willingness to pay for green electricity (WTP) and carbon tax support (TS) were highly correlated with the warm glow measure ($p < 0.001$; Table 7). Both variables correlated also significantly with the variables assessing altruism (AT, AV and EV), and warm glow and altruistic values were significantly correlated, too.

Testing for multicollinearity revealed minimum detection-tolerance values ranging 0.44 to 0.85 and VIF ranging 1.18 to 2.28, confirming that multicollinearity was not an issue. P-P plots of standardized regression residuals were approximately aligned with the diagonal for the TS regressions, indicating normality of the error distribution. For WTP, however, there was a significant deviation. Curve estimation suggested that WTP had a logarithmic rather than a linear relationship with independent variables. Ln(WTP) was therefore used in all analyses. P-P plots of the ln(WTP) regressions confirmed approximately normal distributed standardized regression residuals. Missing values ranged 8 (1.3%) to 36 (6.0%) and were excluded listwise ($N = 600$). Little's MCAR test confirmed randomness of missing values (chi-square = 26.98, $p = 0.17$).

Stepwise multiple regression analyses with the enter method (Table 8) show that warm glow is again the strongest predictor of both proenvironmental behaviors ln(WTP) ($\beta_{WTP8} = 0.50$, $p < 0.001$) and TS ($\beta_{TS8} = 0.48$, $p < 0.001$). The effects of altruistic personality traits and most altruistic value variables turned non-significant, when warm glow was introduced into the model, with the exception of the influence of altruistic self-transcendence values on tax support that remained marginally significant ($\beta_{TS8} = 0.10$, $p = 0.06$). The comparison of the full models ($R^2_{WTP8} = 0.24$ and $R^2_{TS8} = 0.28$) with the respective models excluding warm glow ($R^2_{WTP7} = 0.05$ and $R^2_{TS7} = 0.11$) shows that introducing warm glow increased explained variance significantly ($\Delta F_{WTP} = 43.33$ and $\Delta F_{TS} = 51.96$, $p < 0.001$). Consistent with study 1, these findings support the superior explanatory power of warm glow as an antecedent of environmental behavior, as compared to altruistic values (RQ1).

Further results again confirmed the proposed mediation of altruism effects by warm glow (H2). Table 8 shows that introducing mediator warm glow into the regression together with individual altruism variables (AT, AV, EV; models WTP1 to WTP6 and TS1 to TS6) removed the effects of most altruism measures, with exception of the influences of self-transcendence altruistic values (AV) and environmental-altruistic values (EV) on TS, which were significantly reduced. Bootstrapping mediation analysis with 10,000 bootstrap samples further supported H2, corroborating significant indirect effects of the altruism variables via warm glow on ln(WTP) and TS (Table 9).

Comparison of mean value differences in both subsamples (Table 6) confirmed that warm glow was significantly higher in the green electricity group than in the standard electricity group ($M_{green} = 4.22$, $SD = 0.66$ vs. $M_{standard} = 3.42$, $SD = 1.07$, $F = 122.52$, $p < 0.001$). Green electricity consumers also rated higher in the altruism variables, showed a higher WTP for green electricity and are more prone to support a carbon tax ($p < 0.001$ to $p < 0.05$).

The fact that in study 2 warm glow mostly displaced the effects of altruism (with the exception of the remaining marginal effect of self-transcendence values), whereas in study 1 a significant effect of environmental altruistic values remained, can be explained by study 2's sample composition. In the half of the sample composed

Table 6
Variable mean value comparisons in green and standard electricity sub-samples (Study 2).

	Standard electricity		Green electricity		F
	M	SD	M	SD	
Willingness to pay (WTP)	10.62	21.16	25.58	31.77	46.09***
Carbon tax support (TS)	2.37	1.49	3.17	1.60	38.49***
Altruistic personality traits (AT)	3.18	0.76	3.44	0.80	15.85***
Self-transcendence altruistic values (AV)	5.25	1.50	5.49	1.57	3.74*
Environmental-altruistic values (EV)	4.90	1.77	5.45	1.59	15.45***
Warm glow (WG)	3.42	1.07	4.22	0.66	122.52***

Note. ***$p < 0.001$; **$p < 0.01$; *$p < 0.05$.

Table 7
Variable correlations (Study 2).

	WTP	TS	AT	AV	EV	WG
Willingness to pay (WTP)	1.00					
Carbon tax support (TS)	0.40***	1.00				
Altruistic personality traits (AT)	0.17***	0.18***	1.00			
Self-transcendence altruistic values (AV)	0.12**	0.26***	0.36***	1.00		
Environmental-altruistic values (EV)	0.20***	0.32***	0.33***	0.67***	1.00	
Warm glow (WG)	0.48***	0.51***	0.26***	0.31***	0.49***	1.00
Green choice (GC)	0.41***	0.25***	0.17**	0.08*	0.16**	0.42***

Note. ***p < 0.001; **p < 0.01; *p < 0.05.

Table 8
Effects of altruism, warm glow and prior green electricity choice on log-transformed willingness to pay a price premium for green electricity (ln(WTP), WTP-models) and carbon tax support (TS-models): stepwise multiple regression model comparison (Study 2).

DV	Model	IV					R^2
		AT	AV	EV	GC	WG	
Ln(WTP)	WTP1	0.17***					0.03
	WTP2	0.04				0.48***	0.24
	WTP3		0.12**				0.01
	WTP4		−0.03			0.49***	0.23
	WTP5			0.20***			0.04
	WTP6			-0.04		0.50***	0.23
	WTP7	0.12**	−0.07	0.22***			0.05
	WTP8	0.06	−0.03	−0.04		0.50***	0.24
	WTP9				0.41***		0.17
	WTP10				0.26***	0.37***	0.29
	WTP11	0.03	−0.01	−0.03	0.27***	0.39***	0.30
TS	TS1	0.18***					0.03
	TS2	0.04				0.51***	0.27
	TS3		0.26***				0.07
	TS4		0.11**			0.49***	0.28
	TS5			0.32***			0.10
	TS6			0.09*		0.47***	0.27
	TS7	0.08a	0.06	0.25***			0.11
	TS8	0.02	0.10a	0.01		0.48***	0.28
	TS9				0.25***		0.06
	TS10				0.04	0.50***	0.27
	TS11	0.01	0.10a	0.01	0.03	0.47***	0.28

Note. ***p < 0.001, **p < 0.01, *p < 0.05, ap < 0.10; standardized regression coefficients beta; multiple R-square; N = 600; missing values excluded listwise; DV = dependent variable; IV = independent variable; AT = altruistic personality traits; AV = self-transcendence altruistic values; EV = environmental-altruistic values; GC = green choice (standard electricity = 0; green electricity = 1); WG = warm glow.

of green electricity clients warm glow and behavioral intentions were significantly stronger than in the remaining half of the sample consisting of standard electricity clients. The whole sample had therefore a much higher proportion of proenvironmental individuals experimenting warm glow than a general population sample and warm glow effects were consequently accentuated.

With respect to the proposed reinforcement process (H3), results of study two also confirmed warm glow as a mediator of the effects of prior green electricity choice (GC) on intentions to engage in future proenvironmental behavior. Stepwise regression analyses showed that GC had a significant influence on ln(WTP) (WTP 9) and TS (TS 9) and that this effect was significantly reduced (WTP10) or even removed (TS10), when warm glow was introduced into the regression model (Table 8). This result remained identical when controlling for altruism effects (models WTP11 and TS11).

Mediation analyses of indirect effects by both Sobel tests and bias corrected bootstrap confidence intervals with 10,000 bootstrap samples further confirmed that the effect of prior environmental behavior on future behavior intentions was mediated by warm glow (Table 9). Consistent with H3, a significant indirect effect of GC via warm glow can be observed for ln(WTP) ($b_{WTP\ indirect} = 0.76$, SD = 0.10, 95% Boot CI [0.59, 0.97], Z = 7.31, $p < 0.001$) and TS ($b_{TS\ indirect} = 0.67$, SD = 0.07, 95% Boot CI [0.54, 0.81], Z = 8.31, $p < 0.001$). Results regarding the indirect effect of GC imply that relative to consumers with standard electricity contracts (GC = 0) consumers with green electricity (GC = 1) rated on average 0.76 units higher in their ln(WTP) for green electricity and 0.67 units higher in their likelihood to support the carbon tax. The effect of green electricity choice (GC) is fully mediated for tax support (TS). In the case of ln(WTP), the mediation is partial with a significant direct effect of GC remaining in presence of the mediator, and a 0.61 ratio of indirect to direct effect.

9. Discussion

9.1. Contribution and theoretical implications

Altruism has been proposed in the literature as an antecedent of prosocial and proenvironmental behavior. However, the existence of pure altruism has also been questioned by a number of authors, since empirical evidence for ‘truly’ altruistic behavior has been scarce (e.g., Cialdini et al., 1997). An alternative account for prosocial motivation proposes warm glow as a psychological motive for such behavior, also termed ‘impure altruism’ (Andreoni, 1989, 1990). The aim of the two empirical studies was to analyze the role of warm glow as an antecedent of proenvironmental behavior in a comparative study of the behavioral effects of warm glow and altruistic personality traits and values. The influences of altruism and warm glow have not been previously assessed in the same study and analyzed simultaneously.

The contribution of this research is threefold. First, the study provides evidence to answer the question whether altruistic values or warm glow are stronger drivers of proenvironmental behavior. Findings of both studies regarding the isolated relationship between either altruism or warm glow and proenvironmental behavior are in line with previous research showing that both altruism (e.g., Stern et al., 1993; Stern, Dietz, Abel, Guagnano, & Kalof, 1999) and warm glow (e.g., Kahneman & Knetsch, 1992; Nunes & Schokkaert, 2003; Ritov & Kahneman, 1997) significantly relate to proenvironmental behavior.

However, when simultaneously testing the influence of warm glow and altruistic personality traits and values, the effect of altruism is mostly displaced by warm glow, which indicates that warm glow is the main driver of proenvironmental behavior. This finding questions “reward-free” altruistic motivation and provides evidence for the argumentation of intrinsic emotional reward in proenvironmental behavior (Andreoni, 1989, 1990; Batson, 1987; Cialdini et al., 1997). While we cannot clarify whether proenvironmental behavior is ‘truly’ altruistic or if ‘pure’ altruism actually exists, in our study warm glow was to a greater extent associated with proenvironmental intentions than altruistic traits and values. This finding can explain why individuals lacking

Table 9
Indirect effects of altruism and prior green electricity choice on willingness to pay a price premium for green electricity (ln(WTP) and tax support mediated by warm glow (Study 2).

DV	IV	Indirect Effect (B)	Boot SE	Boot LLCI	Boot ULCI	Z
Ln(WTP)	Altruistic personality traits (AT)	0.38	0.07	0.26	0.52	5.62***
	Self-transcendence altruistic values (AV)	0.24	0.04	0.17	0.32	6.62***
	Environmental-altruistic values (EV)	0.36	0.04	0.28	0.44	8.90***
	Green choice (GC)	0.76	0.10	0.59	0.97	7.31***
TS	Altruistic personality traits (AT)	0.27	0.05	0.19	0.36	5.79***
	Self-transcendence altruistic values (AV)	0.16	0.02	0.11	0.20	6.61***
	Environmental-altruistic values (EV)	0.22	0.02	0.18	0.27	8.70***
	Green choice (GC)	0.67	0.07	0.54	0.81	8.31***

Note. ***$p < 0.001$; 10,000 bootstrap samples for bias corrected bootstrap confidence intervals; DV = dependent variable; IV = independent variable; Boot SE = bootstrap standard error; Boot LLCI = bootstrap lower limit confidence interval; Boot ULCI = bootstrap upper limit confidence interval; WTP = willingness to pay price premium, TS = tax support.

significant altruistic values and personality traits may still engage in seemingly altruistic prosocial and proenvironmental behavior (Batson, 1987; Cialdini et al., 1987, 1997), because they anticipate experiencing positive affective reward. People may incur costs in order to derive intrinsic psychological benefits from experiencing warm glow.

The second contribution of this research is to provide a process explanation for the decreased influence of altruism when warm glow is introduced into the model. Results of both studies support a mediation model with warm glow mediating the effect of altruism on proenvironmental behavior. Instead of an alternative explanation to altruism, warm glow provides a complementary explanation. Indeed, the warm glow mediation further explains altruism effects on behavior. Findings are in line with research showing that altruistic behavior leads to experiencing warm glow (Andreoni, 1990; Kahneman & Knetsch, 1992; Kahneman & Ritov, 1994). Altruistic values seem to increase the experience of warm glow, which in turn mediates altruism effects on proenvironmental behavior.

The direction of the mediation effect is clearly defined theoretically: warm glow mediates the effect of altruism, but altruism does not mediate the effect of warm glow, since the warm glow experience will not change any individual's value structure. Thus, holding altruistic values enhances the experience of warm glow and the anticipated emotional reward drives proenvironmental behavior.

The role of warm glow is however not limited to mediating altruism effects. Introducing warm glow together with altruism explains additional variance of proenvironmental behavior, apart from the indirect effect of altruism. Warm glow explains most altruism effects, but also constitutes a further antecedent of proenvironmental behavior. Altruistic personality traits and values are therefore not a prerequisite for behavioral effects of warm glow. In altruistic individuals, altruistic values and traits may enhance warm glow, whereas individuals lacking significant altruistic traits and values may be motivated by anticipated warm glow alone to engage in proenvironmental behavior.

The third contribution refers to the hypothesized reinforcing mechanism based on the warm glow effect. So far research has studied warm glow as either antecedent or consequence of prosocial and proenvironmental behavior. In line with previous research (e.g., Dunn et al., 2008; Menges et al., 2005; Strahilevitz & Myers, 1998), study 2 confirmed that prior proenvironmental behavior leads to warm glow. However, in addition, the study showed that warm glow mediates the influence of prior green electricity choice on intentions to engage in future proenvironmental behavior. Warm glow can take both the roles of behavioral antecedent and consequence and thus reinforce proenvironmental behavior. As emotional reward derived from contributing to proenvironmental behavior, warm glow experiences are more susceptible to influences and changes than the rather stable altruistic personality traits and values. Experiencing warm glow as a consequence of proenvironmental action may induce learning effects driving future behavior.

9.2. Practical implications

This study has significant implications for marketers and public policy promoting proenvironmental products and behaviors. The good news is that warm glow seems to be a stronger driver for such behavior than altruism. While it is almost impossible to instill altruistic personality traits and values in adults, appealing to warm glow motives and evoking warm glow experiences is more feasible. Individuals lacking strong altruistic motivations can be motivated to engage in proenvironmental behavior by enhancing their awareness of emotional reward. On the other hand, taking into account the proposed indirect effect of altruism on behavior mediated by warm glow, more altruistic individuals will be motivated by warm glow as well. The recommendation derived from this research is that promoters of proenvironmental behaviors, including green product consumption, should focus on enhancing the salience of warm glow derived from such behaviors.

9.3. Limitations and directions for future research

Findings and conclusions should be taken with caution. Although the two samples based on representative population panels contribute to the robustness of results, the cross-sectional character of the data restricts the possibility of causal inferences. Future research should address the proposed relationships with an experimental research method, based on the manipulation of participant's warm glow experiences, for instance, using specific communication claims and framing techniques. Study two approximating a quasi-experimental design based on a sample consisting half of green electricity consumers and half of standard electricity clients, makes a step in this direction, but falls short of an explicit warm glow manipulation. Results showing that warm glow was relatively high also in the standard electricity group may point to a social desirability bias with regard to the warm glow measure, leading non-environmental consumers to report higher scale values. Social desirability bias may therefore have affected results leading to less pronounced differences in warm glow between proenvironmental and non-environmental consumers. Without such a bias, stronger warm glow effects might have been observed. The generalizability of results for general proenvironmental behavior may be limited by the fact that the proenvironmental behaviors tested were not comprehensive and are primarily focused on climate protection and energy consumption behaviors. Future research should address warm glow effects for a broader range of proenvironmental behaviors. In addition, the measurement of emotional and psychological variables still poses important challenges (Bagozzi et al., 1999). The warm glow measure based on operationalizing warm glow as an affective experience should be further validated in different contexts. With regard to the hypothesized mediation model, such a pattern of relations had not been suggested previously. Results concerning the indirect influence of altruism via warm glow must consequently be regarded as exploratory findings. Regarding the behavioral reinforcement process of warm glow, previous green behavior, in addition to its role as behavioral antecedent of warm glow, may also moderate the effect of warm glow on future intentions. Theoretical considerations and further results of the data analysis, while inconclusive and therefore not presented in this paper, point to the possibility of such an effect. Future research should address this possibility.

Some of the theoretical implications of our findings also provide promising avenues for future research, particularly the warm glow mediation of altruism effects. Does the influence of altruistic traits depend on situational affective gratifications? Are there further personality traits involved which determine the individual's susceptibility toward the warm glow effect? Future research should analyze additional traits and aspects of altruism and warm glow together with the influence of moderating variables not yet tested such as, for instance, social norms as well as perceived behavioral efficacy and locus of control. Another avenue for future research is provided by the interaction of warm glow with further psychological mechanisms such as distress reduction that can potentially contribute to explaining seemingly selfless proenvironmental behavior. Besides showing how warm glow mediates the behavioral effects of value-consistent proenvironmental behavior, future studies should therefore also look into the possible mediating effect of negative affect such as guilt or anxiety experienced as consequence of behaving against one's values. Anticipation of psychological punishment may also mediate the effects of altruistic values on proenvironmental behavior.

Acknowledgments

The authors wish to thank the three anonymous reviewers for their constructive comments and suggestions, which contributed significantly to this paper.

This study received financial support from research grants ECO2016-76348-R, GIU11/17, SAIOTEK SA-2013-00351, GIC 10/54-IT 473-10, and FESIDE Foundation.

Appendix 1

Sample characteristics

		Study 1		Study 2	
		Frequency	Percentage	Frequency	Percentage
Gender	Female	299	49.8	323	53.8
	Male	301	50.2	277	46.2
Age	24–29	58	9.67	63	10.5
	30–34	68	11.33	86	14.3
	35–39	79	13.17	80	13.3
	40–44	70	11.67	73	12.2
	45–49	71	11.83	71	11.8
	50–54	59	9.83	54	9.0
	55–59	55	9.17	58	9.7
	60–64	45	7.50	44	7.3
	>65	95	15.83	71	11.8
Household income	$0 - $20,000	42	7.0	22	3.7
	$20,001 - $40,000	76	12.7	86	14.3
	$40,001 - $60,000	96	16.0	79	13.2
	$60,001 - $80,000	71	11.8	77	12.8
	$80,001 - $100,000	73	12.2	84	14.0
	$100,001 - $120,000	58	9.7	53	8.8
	$120,001 - $140,000	37	6.2	54	9.0
	$140,001 - $160,000	34	5.7	26	4.3
	$160,001 - $180,000	13	2.2	11	1.8
	$180,001 - $200,000	12	2.0	20	3.3
	> $200,000	18	3.0	25	4.2
	I prefer not to say	70	11.7	63	10.5
Education	High School - Year 10 qualification	81	13.5	70	11.7
	High School - Year 12 qualification	101	16.8	70	11.7
	Tafe/College	172	28.7	183	30.5
	University - Undergraduate	149	24.8	141	23.5
	University - Postgraduate	79	13.2	116	19.3
	Other	18	3.0	20	3.3
Profession	Full-time employment	287	47.8	293	48.8
	Part-time/casual employment	94	15.7	124	20.7
	Student	11	1.8	16	2.7
	Home duties	58	9.7	52	8.7
	Retired	109	18.2	92	15.3
	Not currently working	41	6.8	23	3.8
	Total	600	100.0	600	100.0

Appendix 2

Excerpt of in-depth interviews

Female, 55: Well I just think that I am doing something like I can't go and be a crew member of Greenpeace or anything like that. Oh well **it's something positive for me**. I know it's very small. Yeah it's positive; it's something **I feel good about**.

Male, 40: Yeah look I feel like that there is a small part of me that is **making a bit of a difference**. That **I feel like I am giving something back to the environment** and maybe I'm just one person but maybe **I am making a very small difference**. (…)Yeah, I feel, ah, not proud, but **I feel a little happy about that decision we made**. So yeah **I feel good about it**. (…) Ah, yeah, it's a very **pleasant feeling**. It's **a feeling that is filling me that you are making a bit of a difference**. (…) **Feeling happy and content**. (…) Probably not a huge amount in the overall scheme of things but I **feel better in my mind and in my heart that I am making a difference**. I guess. That's **pleasing** for me and pleasing for my family (…).

Female, 54: I feel good that I am doing what I can (…). There is a **good feeling that at least I am doing something** (…). Yeah, it's a good feeling or I wouldn't keep doing it.

Female, 59: Well I feel good about it, no **I feel good that I do it**. (…) Well, **I think it's my little contribution**.

Male, 43: I'm very **proud of any positive effects we can have**. I feel, … **I feel very happy to do that**, given that I think it's something … it's very necessary. I think that's why it mainly comes down to that **feeling of pride and satisfaction in doing something that helps**, while knowing it's never going to be quite enough, **it does feel good to know that I can do something**. I also, **I'm really very happy that I can do something**.

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